(i) Consider the first two tosses
If the first toss is T, two consecutive heads must need for either to win, which THH occurs first
If they are HH, then the first later tail gives HHT, so A wins
If they are HT, similarly to the first reason, B will win
The first two tosses are equally likely to be HH, HT, TH, TT

$$P(A \text{ wins}) = \frac{1}{4}(1+0+0+0) = \boxed{\frac{1}{4}}$$

(ii) Consider the first two tosses
If they are HH, then the first later tail gives HHT, so A will win
If they are TT, then the first later head gives TTH, so C will win
If they are HT or TH, only B or D can win
By symmetry, the total probabilities of B and D winning from these two cases are equal

$$\therefore \boxed{P(A) = P(B) = P(C) = P(D) = \frac{1}{4}}$$

(iii) If the first two tosses are TT, then T loops and the first later H gives TTH

$$\boxed{P(C \text{ wins} \mid \text{TT}) = 1}$$

$$p = P(C \mid \text{HT}), \qquad q = P(C \mid \text{TH}), \qquad r = P(C \mid \text{HH})$$

From HT, the next toss is either T, giving TT, or H, giving TH

$$p = \frac{1}{2} \cdot 1 + \frac{1}{2}q$$

From TH, the next toss is either H, giving THH, so B wins, or T, giving HT

$$q = \frac{1}{2} \cdot 0 + \frac{1}{2}p = \frac{1}{2}p$$

From HH, the next toss is either T, giving HT, or H, staying at HH

$$r = \frac{1}{2}p + \frac{1}{2}r$$

$$\implies p = \frac{2}{3}, \qquad q = \frac{1}{3}, \qquad r = \frac{2}{3}$$

The first two tosses are equally likely to be HH, HT, TH, TT

$$
\begin{aligned}
P(C \text{ wins}) &= \frac{1}{4}(r+p+q+1) \\
&= \frac{1}{4}\left(\frac{2}{3}+\frac{2}{3}+\frac{1}{3}+1\right) \\
&= \boxed{\frac{2}{3}}
\end{aligned}
$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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